DELPHI: five people, a few hundred agents

Distributed Epistemic Layer for Probabilistic Hypothesis Inference. The goal: find the fundamental theorems of biology, by extracting candidate laws as claims, proving what can be proved, testing the rest on real organisms, and publishing every step as a nanopublication.



Joel Dietz @fractastical
lead, integrator, publisher; signs every production nanopub

Kris @kganjam
form simulator and queryable open problems

Kavita @kavitakoolkarni-bio
presentation layer: claim graph site, demo, slides

Ashley @ashleycaselli
nanopubs; linking claims to INDRA, GBIF and Wikidata

Amit @xX-its-amit-Xx
genotype–phenotype links, MCP testing, agent comparison

and the agents

Claude builds · GPT-5.6, Gemini 4 and Grok 4.7 assess blind · cloud agents test · Delfie talks to other teams

We arrived with a method already running

fractastical/morpholib (692 commits since August 2025, 561 of them in October): reproductions of published models of biological form, scored by MorphoEval and published as signed nanopubs. Everything this weekend reuses its loop.

127
MODEL REPRODUCTIONS, THOMPSON 1917 TO 2026

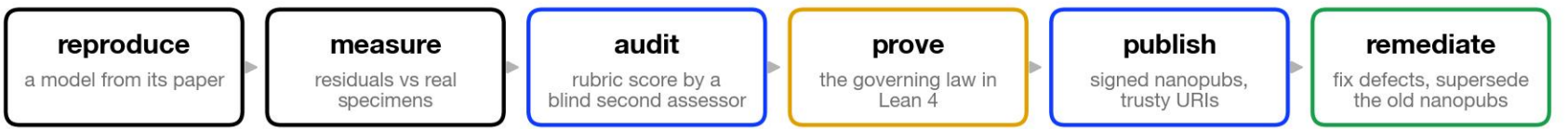
1,080
DATASETS VENDORED WITH PROVENANCE

201
NANOPUBS ON THE PRODUCTION NETWORK

25
LAWS MACHINE-CHECKED IN LEAN 4, 0 SORRY

8,740
MORPHOEVAL AUDIT FILES

THE MORPHOEVAL LOOP, THE PATTERN REUSED ALL WEEKEND

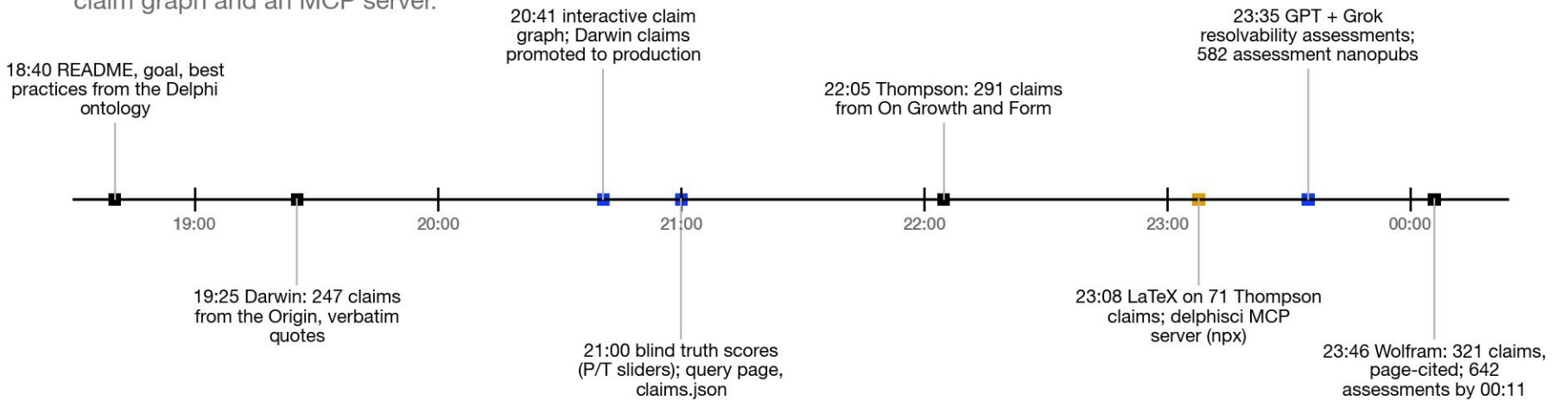


Worked example: Runions 2005 leaf venation scored 40.9/100 on pass 1; the blind second assessor found five defects; pass 2 fixed them and scored 91.3, superseding the pass-1 nanopubs. Thompson's transformations got 49 blind, pre-registered predictions tested on specimens and museum lithographs. The problem graph (solved / partial / open / refuted) is the ancestor of this weekend's claim graph.

Also carried in: the Project Delphi ontology (fractastical/project-delphi-ontology) that every nanopub here is written against, and the morphoswarm Frontier node at frontier.morphoswarm.ai.

Friday: three books become 859 claims

First commit 18:40. By midnight: Darwin, Thompson and Wolfram extracted with verbatim quotes, every claim scored blind by two models for whether open data could settle it, 1,500+ nanopubs on the test registry, a claim graph and an MCP server.



859
CLAIMS BY MIDNIGHT

1,224
ASSESSMENT NANOPUBS, TWO BLIND MODELS

71
CLAIMS WITH A FORMAL STATEMENT

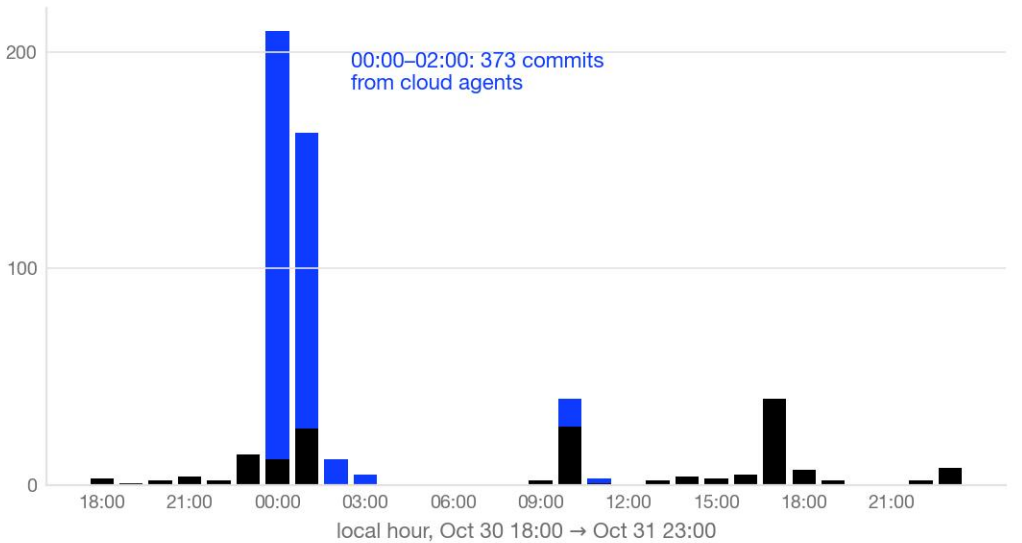
1
MCP SERVER, RUNNABLE BY ANY AGENT

26
COMMITS BY ONE PERSON AND ONE BUILDER AGENT

Overnight: 19 cloud agents on 19 branches

At 00:16 two specs were written: how to test a claim against data (pre-register pass/fail criteria, then fetch) and how to vendor organism reference data. Then 13 claim-test agents and 6 organism agents ran in parallel while the team slept.

COMMITS PER HOUR · BLACK = PEOPLE AND THE BUILDER AGENT · BLUE = CLOUD AGENTS



13 claim-test agents

4 Darwin, 4 Thompson, 5 Wolfram. Each pre-registers a result.json with hypothesis and pass/fail criteria, then downloads the data (PBDB, GBIF, Open Tree, WCVP, OEIS, NHANES, FishBase …) and writes the verdict. 129 results merged at 10:02.



VERDICTS, ALL 161 TESTED CLAIMS SO FAR; 129 RUNS FROM THIS NIGHT

6 organism agents

Every animal taxon Darwin and Thompson mention, resolved to GBIF / NCBI / Wikidata / WoRMS / PBDB with occurrence counts, genomes and trait databases. 782 taxa, 3,340 located mentions, 1,372 dataset records, merged at 01:40.

00:32 the MCP server goes to Fly.io over Streamable HTTP, deployed from main by GitHub Actions. 01:52 Amit pre-registers and runs the four-arm agent comparison (slide 6).

Saturday: merge, verify, widen, publish

The morning merged the overnight branches and turned the agents on their own outputs; the afternoon added Levin, Lean proofs, Aristotle and Nebius, and published Thompson and Wolfram to production.

09:40–10:05	**Merge and show**	19 branches merged: 129 verdicts, 938 taxa; organisms in the MCP; README gets the snapshot, commit-history and organisms GIFs.
10:05–11:02	**Agents check the agents**	Dataset-reliability pass: 4 agents re-check 1,679 organism dataset records against NCBI, GBIF, PBDB/OBIS and 13 shared databases (1,603 reliable, 72 caveats, 4 needs repair).
10:38	**Nebius: cell geometry**	Cellpose-SAM segments micrographs of five tissues to test Thompson's 120° junctions and hexagon claims; the segmenter-bias control later cuts ch07-04 from 0.60 to 0.35.
13:43	**Levin and Lean**	Michael Levin joins: 16 CC-BY papers, 310 claims, all scored. 47 Thompson claims machine-checked in Lean 4 (no sorry), 71 by now with Darwin's. Agent forum, Discord agent, executable-claims runner.
14:07	**Production**	Thompson's 291 and Wolfram's 321 claims with their assessments signed with the operator's ORCID key and pushed to the production registry: 372 → 880 live.
14:14–15:45	**Levin tested**	Pre-registered tests of Levin's claims: GA competency replication, the TAME gene-overlap prediction (two halves), and three protein-structure tests folded on Nebius.
16:03	**Restructure; Aristotle**	claim-sources / test-data / scripts / plugins. Aristotle's History of Animals, Parts and Generation: 512 claims from ten extractors. Team page, Fittest plugin, forum MCP tools.

Evening: test, grade, measure, connect

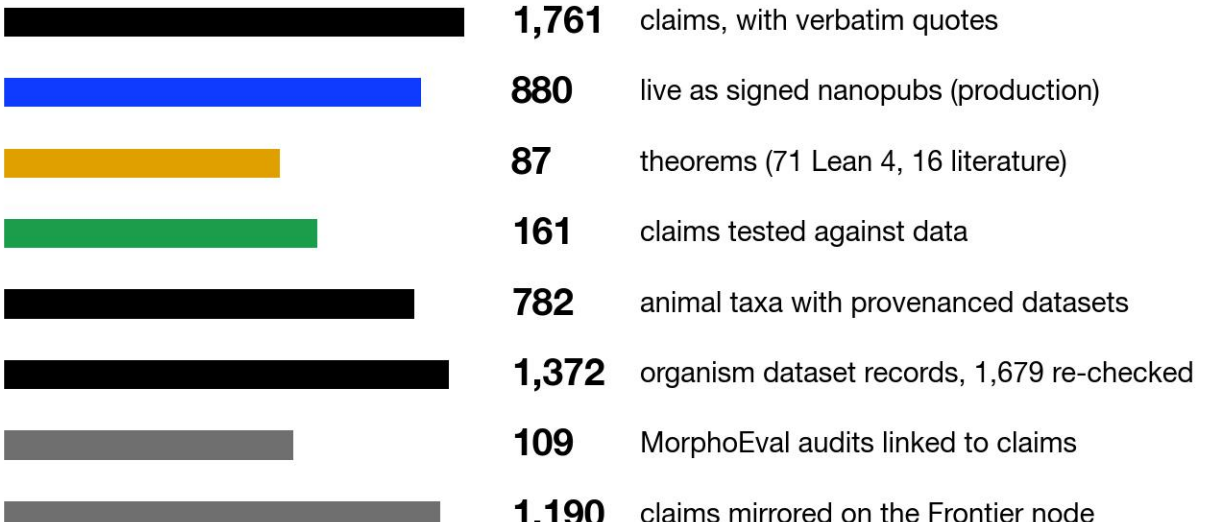
Twenty-five more tests in two hours, then outward: morpholib's audits and the Frontier node joined to the claim graph, three outside models scored the repo, and the swarm was measured against single agents.

17:10–18:58	**25 tests in two hours**	Darwin island biogeography on WCVP + Open Tree (two refutes, two mixed), Wolfram randomness batteries, Thompson heart and fish-shape claims (refutes), Levin keratocytes, nicotine, planarian heads (supports). More have landed since: 161 claims tested now.
17:50	**morpholib ⇄ delphi**	delphi_morpholib: 109 production MorphoEval audits linked to the claims they test (35) or model (69), served from the hosted MCP.
18:14	**Blind self-assessment**	Frozen evidence pack; GPT-5.6 Sol, Gemini 4 Argon and Grok 4.7 score the rubric blind: 57–60/100, published as nanopubs with each assessor's single best improvement.
18:48	**Frontier node**	frontier.morphoswarm.ai now holds 1,326 claims, 1,190 of them delphisci claims by their w3id IRIs, plus 291 problems, 629 artifact declarations and 8 formal-evidence records.
19:00–23:00	**Swarm vs single agent**	Aristotle Book VI: the swarm against Sonnet, Opus and GPT-5.6 each running the whole chain alone. Fittest-style routing on assessor disagreement: 0.83 → 0.935 when trusted. Forum leaderboard; 4 open requests for other teams.
SUN 00:59	**Four arms, imported**	Amit's pre-registered runs (Sat 01:52): 20 claims, one test plan each, same model and budget per arm. Verified datasets of 20: single 13, isolated 17, centralized 19, decentralized 18; with web tools required 12 / 16 / 18 / 17. Every collective arm beats single by 4+.

Meanwhile, outside the repo: Delfie (our agent) in the SentiLab room and on Moltbook; the Fittest MCP pilot and a calibration finding sent back; two upstream nanopub bugs reported (#37); SentiLab's collagen run given its physics anchors.

Thirty hours, three graphs, one method

Everything below is a file in one of two repos or a record on the Frontier node. Mapping it onto the judging criteria is the next step; the material is here.



1,761	claims, with verbatim quotes
880	live as signed nanopubs (production)
87	theorems (71 Lean 4, 16 literature)
161	claims tested against data
782	animal taxa with provenanced datasets
1,372	organism dataset records, 1,679 re-checked
109	MorphoEval audits linked to claims
1,190	claims mirrored on the Frontier node

WHO DID IT

5	people
101	local agent chats
262	subagents
19	cloud agents
3	blind assessor providers
2+1	repos + frontier node

BAR LENGTH IS LOGARITHMIC · COUNTS FROM DOCS/CLAIMS.JSON, TEST-DATA/, MORPHOLIB LINKS.JSON AND FRONTIER_CATALOGUE